
Tyler Drake

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Audio Software Developer / Audio Software Engineer

Software development leader with 10+ years building real-time, high-performance systems. Experienced in C++, Java, and cross-platform development, and an active recording artist passionate about advancing digital audio technology.

Work Experience

CHARLES SCHWAB • 06/2018 - Present

Software Development Manager

Denver, Colorado, United States

- Led Agile development teams through sprint planning, retrospectives, and backlog refinement, delivering complex software on time while ensuring alignment with strategic product goals and user experience needs.
- Owned and optimized JIRA Advanced Roadmaps and Confluence artifacts, giving teams clear visibility into release schedules, capacity, and long-term initiatives—streamlining delivery of complex, high-performance software
- Built automation tooling for release processes, integrating JIRA workflows, documentation, and deployment pipelines to eliminate errors, accelerate handoffs, and ensure reliable production delivery.
- Designed and deployed a modern real-time verification backend from scratch, processing sensitive client data securely and at scale to detect and prevent identity theft and fraud.
- Championed JIRA best practices by creating custom boards, filters, and automation rules, boosting team velocity and ensuring clarity across complex software projects.
- Collaborated cross-functionally with product and business owners to translate stakeholder needs into actionable user stories, technical documentation, and implementation plans
- Built, enhanced, and maintained CI/CD pipelines for 11 mission-critical API and messaging applications across 5 isolated environments (dev, QA, performance, UAT, production), ensuring consistent, high-quality delivery at scale
- Developed advanced Splunk dashboards, alerts, and queries delivering real-time insights into fraud detection, identity data, and system health—including performance, load, exceptions, and transaction flow.
- Managed and migrated a diverse enterprise tech stack including Java 17, Spring Boot 3, Maven, RabbitMQ, MongoDB, SQL, JIRA, Confluence, Git, Bamboo, and Pivotal Cloud Foundry, and Github Copilot ensuring stability and modernized delivery pipelines.
- Ensured all applications remained in compliance with enterprise security standards by integrating Veracode and Black Duck scans, remediating vulnerabilities, and maintaining secure software delivery.

IBM • 03/2016 - 06/2018

Developer

Monroe, Louisiana

- Creating automated processes for web service frameworks with REST and SOAP standards by providing code-writing-code software. Supplementing it with hotfixes, brand new features, and new methods for filing .pom/bean snippets, schema/domain .xsd/.wsdl files, .java files, and comment stamps into developers local workspace.
- Resolved a major pain point by reducing web service timeouts by over 1000%, improving reliability for critical customer calls through data-driven dependency analysis.
- Documented and presented high-level architecture for DVAR-OM web services, outlining system layers, integrations, schemas, and workflows to guide development and ensure scalable, reliable delivery.
- Technical lead, covering development, maintenance, and support for a suite of web applications vital to customer order flow and the provisioning of telecommunication service lines.
- Partnered with architects, managers, systems analysts, product owners, scrum masters, developers, testers, dbas, product owner partners, and directors to drive business.

GREAT-WEST FINANCIAL • 04/2015 - 11/2015

Developer

Greenwood Village

- Utilized Thunderhead software and existing integrated Business Object to code the front-end of database-driven retirement letters based on business requirements.
- Completed over 75 independent projects during the duration of tenure in a waterfall development environment. Assembled documents by creating projects within Thunderhead's Business Content studio, linked individual projects to a business object based on an XML schema and combined static fragments, links to shared content, images, tables, conditional if-then and for-loop statements, and dynamic scripts.

STONEBRIDGE COMPANIES • 11/2013 - 03/2015

Server

Greenwood Village

IVY AT THE GLENN • 05/2012 - 11/2013

Server

Littleton, Colorado

CHARLES SCHWAB • 02/2010 - 11/2011

Voice Analyst

Greenwood Village, Colorado

- Developed and enhanced the front-end of Schwab's Interactive Voice Response (IVR) system using VXML, delivering intuitive, reliable voice menus aligned with business requirements.
- Pioneered and developed a prototype voice application for Schwab's Retirement Plan Services using JSP, VXML, WebSphere, and Oracle SQL. The success of this model secured funding for a large-scale IVR system rollout.
- Produced and demonstrated another prototype iPhone application using Cocoa to exhibit the ability to use voice response in iPhone/iOS applications.
- Connected back-end data through front-end VXML to the customer using SOAP web services.

HEWLETT-PACKARD • 06/2009 - 06/2010

Intern

Fort Collins, Colorado Area

- Implemented automated JUnit unit testing tool to validate both pre-written unit tests and source code build using Cruise Control software, Microsoft Visual Studio 2005, Tortoise SVN, and XML scripts.

Projects

BLOOM • 01/2024 - Present

Tyler James Drake and Hyperthreat Sound

Recording my debut album, teaser at <https://tylerjamesdrake.bandcamp.com/music>

Responsible for writing, arranging, recording, and producing all tracks in my personal studio and lifting the conception to the professional audio environment. Hands-on involvement across DAW workflows (Pro Tools + DadMan), plugin integration, guitar/bass/drums/vocals/horns tracking, and mixing. This project informs my deep understanding of creative pain points and opportunities in modern recording environments.

DAWs & Interfaces: Pro Tools 2025 (Windows & macOS, MTRX Studio interface), Melodyne, Waves, UAD, VocAlign.

Studio Hardware: Dangerous Monitor ST, API 512C/527/550A channel strips, Rupert Neve 5211, SSL XLogic Alpha VHD,

Warm Audio WA-273, Vintech Model 473, Little Labs Redcloud balanced attenuators, Switchcraft 6425 patchbays, Roland

Integra-7 sound module.

Guitars & Basses: Fender Stratocaster, Epiphone Les Paul, Yamaha FG-04-LTD acoustic, Yamaha TRBX174EW bass, BC

Rich Shredzilla custom AI designed guitar (DiMarzio pickups).

Pedals & Effects: Line 6 HX Stomp, Ibanez Tube Screamer, BOSS Loop Station, Electro-Harmonix Holy Grail, Tech21

California preamp, multiple delay/modulation pedals.

SELF REMEDY

Self Remedy business functionality will allow clients to self resolve verification related issues by way of identification document verification.

ENHANCED CLIENT VERIFICATION

The ECV Platform provides real-time identity verification and risk assessment services of Schwab clients across multiple lines of business and channels. The platform will consist of a collection of enterprise ready utility services that may be used across Schwab domains for both new and existing clients.

CRIS TO ENSEMBLE

Billing system merger.

FLOW THROUGH SOLUTIONS

Flow Through Solutions is a web application suite architected to provide smooth customer order flow and provisioning.

Education

COLORADO STATE UNIVERSITY

Bachelor's degree in Computer Science • 01/2006 - 12/2009

HERITAGE HIGH SCHOOL

Graduated with High Honors • GPA: 4.0.6 • 01/2001 - 12/2005

Certifications

CSRF 101: CROSS SITE REQUEST FORGERY FOR EVERYONE • 06/2021 - Present

SAFECode.org

PYTHON • 09/2020 - Present

HackerRank

JAVA • 09/2020 - Present

HackerRank

PROBLEM SOLVING • 09/2020 - Present

HackerRank

IBM DIGITAL LEARNING TRANSFORMATION • 05/2018 - Present

IBM

IBM CLOUD GARAGE METHOD EXPLORER • 03/2018 - Present

IBM

IBM BLOCKCHAIN ESSENTIALS • 02/2018 - Present

IBM

COGNITIVE PRACTITIONER • 01/2018 - Present

IBM

IBM DESIGN THINKING PRACTITIONER • 11/2017 - Present

IBM

SPLUNK CERTIFIED USER • 06/2016 - Present

Splunk

JIRA PROJECT MANAGMENT

JIRA

Awards & Scholarships

MANAGER'S CHOICE AWARD • 11/2017

Mark Hopkins

SERVICE EXCELLENCE AWARD • 10/2016

Brien Louque

Volunteering & Leadership

UNITED WAY OF NORTHEAST LOUISIANA • 11/2016 - 03/2018

Reader

Volunteer reading opportunity for elementary students.

ST. ANDREW'S UNITED METHODIST CHURCH • 07/2006 - 07/2006

Volunteer

Aided in construction of a new school building in Belize.

Skills

- | | | | | |
|-----------------------|--------------------|-------------------------|------------------------|----------------------|
| - Agile Methodologies | - Leadership | - Spring Framework | - Waves | - multi-instrumental |
| - Automation | - Lisp | - SQL | - UAD | performance (guitar) |
| - Bash | - Mac OS | - Test Driven | - VocAlign | - bass |
| - BASIC | - Maven | Development | - plugin integration | - vocals |
| - CLI | - MongoDB | - Unix | - MIDI sequencing (| - drums |
| - Continuous | - Multithreading | - VXML | Akai MPK Mini) | - keys) |
| Integration | - MVC Architecture | - Windows | - Roland Integra-7 | - mixing & mastering |
| - CSS | - OPENQASM | - XML | - patchbay routing | - gain staging |
| - Customer Service | - Pro Tools | - Pro Tools 2025 (| - analog summing (| - signal flow |
| - Debugging Code | - Python | Windows & macOS) | 5059 Satellite) | optimization |
| - Git | - Quantum | - Melodyne | - balanced attenuation | - creative workflow |
| - HTML5 | - RabbitMQ | - DAW workflows | (Little Labs | design |
| - Hyperledge | - Rally | - plugin integration | Redcloud) | - live performance |
| Composer Modeling | - Representational | - MIDI sequencing (| - guitar/bass/drum | experience |
| Language | State Transfer (| Akai MPK Mini) | recording & editing | - music distribution |
| - Java | REST) | - virtual instruments (| - Manley VOXBOX | - Apple Music |
| - JavaScript | - RESTful | Roland Integra-7) | - Songwriting | - Bandcamp |
| - JUnit | WebServices | - MTRX Studio) | - arranging | - Audius |
| | | - Melodyne | | - Spotify |